Syllabus 9758

Statistics Summary

Maximus Pung

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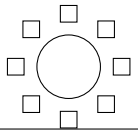
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1 Permutations and Combinations

1.1 Counting Principles and Formulae

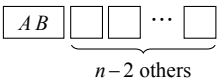
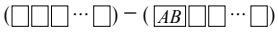
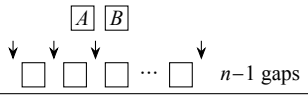
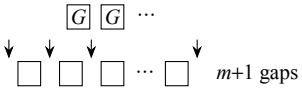
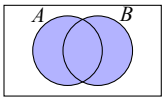
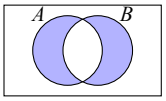
	Addition principle	Multiplication principle
Use when	either event A **or** B (cases)	stage 1 **and** stage 2 (steps)
Condition	cases cannot happen together	stages are done one after another
Total	$m+n$	$m\times n$

Situation	Formula	Visual	Note
n distinct objects in a row	$n!$	$\boxed{n}\boxed{n-1}\boxed{n-2}\cdots\boxed{1}$	$0!=1$
r of n distinct objects, in order	${}^nC_r\times r!=\dfrac{n!}{(n-r)!}$	$\boxed{n}\boxed{n-1}\cdots\boxed{n-r+1}$	order matters
r of n distinct objects, no order	${}^nC_r=\dfrac{n!}{r!\,(n-r)!}$	$\underbrace{\bullet\bullet\bullet\bullet}_{\text{pick } r}\bullet\bullet\bullet\cdots\bullet$	${}^nC_r={}^nC_{n-r}$
r places, each of n choices, repeats allowed	n^r	$\boxed{n}\boxed{n}\boxed{n}\cdots\boxed{n}$	objects **can** be reused
n objects, p alike of one kind, q alike of another	$\dfrac{n!}{p!\,q!}$	$\underbrace{\boxed{a}\cdots\boxed{a}}_{p\text{ times}}\underbrace{\boxed{b}\cdots\boxed{b}}_{q\text{ times}}$	divide by $k!$ for each k identical
n distinct objects in a circle	$\dfrac{n!}{n}=(n-1)!$		rotations are the same

GC: nPr, nCr: MATH → PRB. Type n first, then the command, then r. ! is in the same menu.

> **Note**
>
> Order matters ⇒ select then arrange: ${}^nC_r\times r!$

1.2 Arrangements in a Line

Condition	Method	Count	Visual
Fixed position (e.g. A first; number is even; first digit $\neq 0$)	fill the restricted places **first**, then the rest	$\underline{k} \times _ \times \cdots$	
A, B **together**	Grouping method: group A and B together, then arrange the blocks; arrange inside the block.	$(n-1)! \times 2!$	
A, B **separated** (complement)	total − together	$n! - (n-1)!\,2!$	
A, B **separated** (slot in)	slot in: arrange the other $n-2$; choose 2 of the $n-1$ gaps for A, B; arrange A, B	$(n-2)! \times {}^{n-1}\mathrm{C}_2 \times 2!$	
No two of a group adjacent	**slot in method**: arrange the other m; choose k of the $m+1$ gaps for the group; arrange the group	$m! \times {}^{m+1}\mathrm{C}_k \times k!$	
"at least one"	complement: total − none		
A **or** B (or both)	add, then subtract the overlap once	$n(A)+n(B)-n(A\cap B)$	
A **or** B but not both	add, then subtract the overlap twice	$n(A)+n(B)-2n(A\cap B)$	
Several cases (e.g. last digit 0 or not)	case table: count each case, then **add**		

Example 1.1

Find the number of arrangements of the letters in the word STATISTICS.

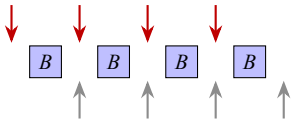
$$\text{S}\times 3,\ \text{T}\times 3,\ \text{I}\times 2$$

$$\text{total} = \frac{10!}{3!\,3!\,2!}$$

$$= 50\,400$$

Example 1.2

Find the number of ways 4 boys and 4 girls can be arranged in a line such that boys and girls alternate.

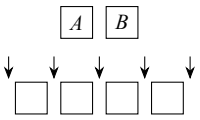


$$\text{total} = 2\times 4!\times 4!$$

$$= 1152$$

Example 1.3

How many ways are there to arrange 6 people in a line with A and B separated?

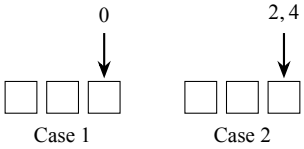


$$\text{total} = 4!\times {}^5\text{C}_2\times 2!$$

$$= 480$$

Example 1.4

Even 3-digit numbers from 0, 1, 2, 3, 4, 5, no repeats



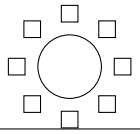
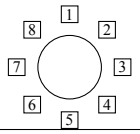
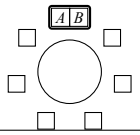
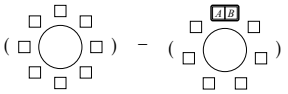
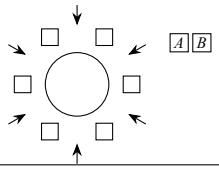
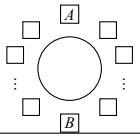
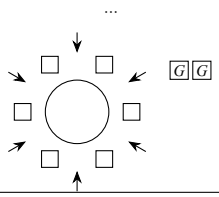
$$\text{last } 0 = 5\times 4 = 20$$

$$\text{last } 2 \text{ or } 4 = 2\times 4\times 4 = 32$$

$$\therefore \text{total} = 20+32$$

$$= 52$$

1.3 Circular Arrangements

Situation	Method	Count	Visual
n people round a table		$(n-1)!$	
seats **numbered / distinguishable**	ignore the numbered seats and do everything as per normal, then multiply by the number of seats at the end	$(n-1)!\times n$	
A, B **together**	grouping, then circular arrangement of the blocks	$(n-2)!\times 2!$	
A, B **separated** (complement)	total − together	$(n-1)!-(n-2)!\times 2!$	
A, B **separated** (slot in)	slot in: arrange the other $n-2$ in a circle; choose 2 of the $n-2$ gaps for A, B; arrange A, B	$(n-3)!\times {}^{n-2}\mathrm{C}_2\times 2!$	
A, B in fixed relation (e.g. A opposite B)	fix A; B has 1 place; arrange the rest	$(n-2)!$	
no two of k people adjacent	**slot in method**: arrange the other m in a circle; choose k of the m gaps; arrange the k people	$(m-1)!\times {}^{m}\mathrm{C}_k\times k!$	

1.4 Selections and Groups

Situation	Method	Count
choose r of n, with a condition (e.g. ≥ 1 woman)	${}^nC_r$; cases (add), or complement	total − none
choose, **then** arrange	select, then multiply by $r!$ (or by the arrangements)	${}^nC_r \times r!$
"at least one of **each** kind"	consider cases Do **not** choose one of each first, then mix the rest together and choose the remaining (overcounts)	add cases
forming words / codes with **repeated** letters	cases by the pattern of repeats (all different; one pair; …): choose the letters, then arrange $\frac{r!}{p!\,q!}$	add the cases
groups of **different** sizes / labelled groups	choose group by group; named teams, different tasks or sizes: **no** division	${}^nC_a \, {}^{n-a}C_b \cdots$
k groups of the **same** size, unlabelled	as above, then $\div k!$ (identical groups of equal size)	$\frac{{}^nC_a \, {}^{n-a}C_a \cdots}{k!}$

Example 1.5

8 people into two groups of 4

$$\text{unlabelled} = \frac{{}^8C_4}{2!} = 35$$

$$\text{teams } X, Y = {}^8C_4 = 70$$

Example 1.6

3-letter codes from MISSISSIPPI

$$\text{all different} = {}^4C_3 \times 3! = 24$$

$$\text{one pair} = (3 \times 3) \times \frac{3!}{2!} = 27$$

$$\text{all alike} = 2$$

$$\therefore \text{total} = 53$$

Example 1.7

5 from 3 red, 4 green, 5 blue; every colour used

$$(3, 1, 1) = {}^3C_3 \times {}^4C_1 \times {}^5C_1 = 20$$

$$(1, 3, 1) = {}^3C_1 \times {}^4C_3 \times {}^5C_1 = 60$$

$$(1, 1, 3) = {}^3C_1 \times {}^4C_1 \times {}^5C_3 = 120$$

$$(2, 2, 1) = {}^3C_2 \times {}^4C_2 \times {}^5C_1 = 90$$

$$(2, 1, 2) = {}^3C_2 \times {}^4C_1 \times {}^5C_2 = 120$$

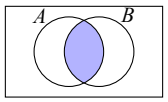
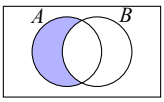
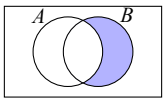
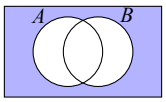
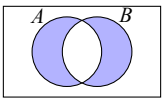
$$(1, 2, 2) = {}^3C_1 \times {}^4C_2 \times {}^5C_2 = 180$$

$$\therefore \text{ways} = 20 + 60 + 120 + 90 + 120 + 180 = 590$$

2 Probability

2.1 Basic Rules

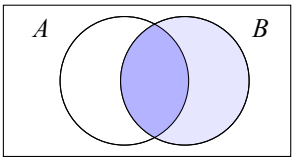
Result	Formula	Note
Equally likely outcomes	$\mathrm{P}(A)=\frac{n(A)}{n(S)}$	$0\leqslant \mathrm{P}(A)\leqslant 1$
Complement	$\mathrm{P}(A')=1-\mathrm{P}(A)$	"at least one" = 1 – none
Addition rule	$\mathrm{P}(A\cup B)=\mathrm{P}(A)+\mathrm{P}(B)-\mathrm{P}(A\cap B)$	$A\cup B$: A or B (or both)
Mutually exclusive	$\mathrm{P}(A\cap B)=0$	so $\mathrm{P}(A\cup B)=\mathrm{P}(A)+\mathrm{P}(B)$
Exhaustive	$\mathrm{P}(A\cup B)=1$	

 $A\cap B$	 $A\cap B'$	 $A'\cap B$	 $A'\cap B'=(A\cup B)'$	 exactly one
$\mathrm{P}(A\cap B)$	$\mathrm{P}(A)-\mathrm{P}(A\cap B)$	$\mathrm{P}(B)-\mathrm{P}(A\cap B)$	$1-\mathrm{P}(A\cup B)$	$\mathrm{P}(A)+\mathrm{P}(B)-2\mathrm{P}(A\cap B)$

The four regions add up to 1. $(A\cup B)'=A'\cap B'$ and $(A\cap B)'=A'\cup B'$. None of these probability results is in MF27.

2.2 Conditional Probability

Result	Formula	Note
Conditional probability	$\mathrm{P}(A\mid B)=\frac{\mathrm{P}(A\cap B)}{\mathrm{P}(B)}$	probability of A given B has happened; $\mathrm{P}(B)\neq 0$
Multiplication rule	$\mathrm{P}(A\cap B)=\mathrm{P}(A\mid B)\mathrm{P}(B)=\mathrm{P}(B\mid A)\mathrm{P}(A)$	used along a tree branch
Complement given B	$\mathrm{P}(A'\mid B)=1-\mathrm{P}(A\mid B)$	
Warning	$\mathrm{P}(A\mid B)\neq \mathrm{P}(B\mid A)$	different denominators

reduced sample space: 

$$\mathrm{P}(A\mid B)=\frac{\text{shaded}}{\text{whole circle } B}$$

2.3 Independent and Mutually Exclusive Events

	Independent	Mutually exclusive
Meaning	one event does not change the probability of the other	cannot both happen
Test	• $\mathrm{P}(A\cap B)=\mathrm{P}(A)\mathrm{P}(B)$ • $\mathrm{P}(A\mid B)=\mathrm{P}(A)$ • $\mathrm{P}(B\mid A)=\mathrm{P}(B)$	• $\mathrm{P}(A\cap B)=0$ • $\mathrm{P}(A\cup B)=\mathrm{P}(A)+\mathrm{P}(B)$
Venn	circles overlap, $\mathrm{P}(A\cap B)=\mathrm{P}(A)\mathrm{P}(B)$	circles do not overlap

Note

- If A, B are independent, so are A and B', A' and B, A' and B'.
- If $\mathrm{P}(A)>0$ and $\mathrm{P}(B)>0$, events cannot be both independent and mutually exclusive: $\mathrm{P}(A\cap B)=0\neq\mathrm{P}(A)\mathrm{P}(B)$.
- To show independence, show $\mathrm{P}(A\cap B)=\mathrm{P}(A)\mathrm{P}(B)$; *do not* assume it from the wording.
- Independence cannot be detected from the shape of the Venn diagram: it generally looks the same as a Venn diagram in which A and B are not independent.

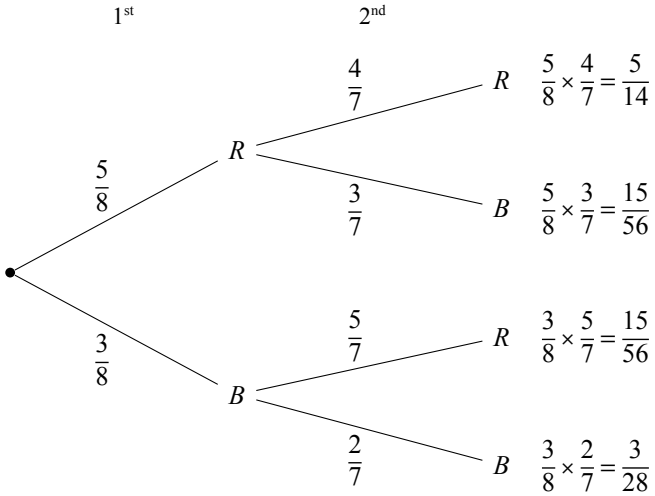
Example 2.1

Given that A is independent of B, prove that A' is independent of B'.

$$\begin{aligned}\mathrm{P}(A'\cap B') &= 1-\mathrm{P}(A\cup B)\\ &= 1-\mathrm{P}(A)-\mathrm{P}(B)+\mathrm{P}(A\cap B)\\ \because\ \mathrm{P}(A\cap B) &= \mathrm{P}(A)\mathrm{P}(B)\\ \therefore\ \mathrm{P}(A'\cap B') &= 1-\mathrm{P}(A)-\mathrm{P}(B)+\mathrm{P}(A)\mathrm{P}(B)\\ &= (1-\mathrm{P}(A))(1-\mathrm{P}(B))\\ &= \mathrm{P}(A')\mathrm{P}(B')\\ \therefore\ A',\ B' &\text{ are independent}\end{aligned}$$

2.4 Tree Diagrams

Multiply along branches (∩); add across leaves (∪). Branches from one node sum to 1. Without replacement: probabilities in the second stage **change**.



Bag: 5 red, 3 blue; 2 drawn without replacement.

$$\begin{aligned}\mathrm{P}(\text{different colours}) &= \frac{5}{8}\times\frac{3}{7}+\frac{3}{8}\times\frac{5}{7}\\ &= \frac{15}{56}+\frac{15}{56}\\ &= \frac{15}{28}\end{aligned}$$

$$\begin{aligned}\mathrm{P}(1\text{st } R \mid 2\text{nd } B) &= \frac{\mathrm{P}(1\text{st } R \text{ and } 2\text{nd } B)}{\mathrm{P}(2\text{nd } B)}\\ &= \frac{\mathrm{P}(R,B)}{\mathrm{P}(R,B \text{ or } B,B)}\\ &= \frac{\frac{15}{56}}{\frac{15}{56}+\frac{6}{56}}\\ &= \frac{5}{7}\end{aligned}$$

2.5 Tables of Outcomes and Counting Methods

+	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

two dice: $n(S)=36$

$$\mathrm{P}(\text{sum}=7)=\frac{6}{36}=\frac{1}{6}$$

$$\mathrm{P}(\text{sum}\geqslant 10)=\frac{6}{36}=\frac{1}{6}$$

$$\mathrm{P}(\text{sum}=7 \mid \text{a 3 shown})=\frac{2}{11}$$

Example 2.2

There are 6 men and 10 women. 5 people are chosen. Find the probability that the group has 3 men.

(a)

$$\begin{aligned}\mathrm{P}(3M, 2W) &= \frac{6}{16}\times\frac{5}{15}\times\frac{4}{14}\times\frac{10}{13}\times\frac{9}{12}\times\frac{5!}{2!\,3!}\\ &= \frac{15}{728}\times 10\\ &= \frac{75}{364}\end{aligned}$$

(b)

$$\begin{aligned}\mathrm{P}(3 \text{ men}) &= \frac{{}^{6}\mathrm{C}_3\times{}^{10}\mathrm{C}_2}{{}^{16}\mathrm{C}_5}\\ &= \frac{20\times 45}{4368}\\ &= \frac{75}{364}\end{aligned}$$

2.6 Further Question Types

Type	Method
Range of $\mathrm{P}(A\cap B)$	Label $\mathrm{P}(A\cap B)=x$, then label all other areas in terms of x. Let each area $\geqslant 0$ and solve.
Games: "first to win", taking turns	list the winning sequences; the game can go on forever; the probabilities form a **GP**; $S_\infty = \frac{a}{1-r}$ $(\lvert r\rvert<1)$
Venn diagram with unknowns (x, y)	label the regions (2 or 3 events) with x, y, …; regions add to 1; use some other info in the equation to link x and y; every region $\geqslant 0$ gives the greatest / least value

Example 2.3

$\mathrm{P}(A)=0.45$, $\mathrm{P}(B)=0.7$. Find the range of values of $\mathrm{P}(A\cap B)$.

let $\mathrm{P}(A\cap B)=x$

$\mathrm{P}(A\cap B')=0.45-x$

$\mathrm{P}(A'\cap B)=0.7-x$

$\mathrm{P}(A'\cap B')=1-(0.45-x)-(0.7-x)-x$

$=x-0.15$

$\because$ each area $\geqslant 0$

$\therefore 0.15\leqslant \mathrm{P}(A\cap B)\leqslant 0.45$

Example 2.4

A, B take turns to roll a die, A first; first six wins

$$\begin{aligned}\mathrm{P}(A \text{ wins}) &= \frac{1}{6}+\left(\frac{5}{6}\right)^2\frac{1}{6}+\left(\frac{5}{6}\right)^4\frac{1}{6}+\cdots\\ &= \frac{1/6}{1-25/36}\\ &= \frac{6}{11}\end{aligned}$$

3 Discrete Random Variables

3.1 Probability Distribution

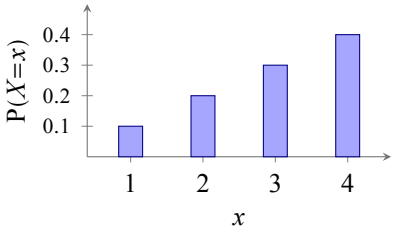
Item	Result	Note
Discrete r.v. X	takes separate values $x_1, x_2, \ldots$ (counts)	e.g. number of heads
Probability distribution	table of x against $\mathrm{P}(X=x)$	
Conditions	$0 \leqslant \mathrm{P}(X=x) \leqslant 1$ and $\sum_{\text{all } x} \mathrm{P}(X=x) = 1$	use to find an unknown constant

x	1	2	3	4
$\mathrm{P}(X=x)$	k	$2k$	$3k$	$4k$

$$\sum \mathrm{P}(X=x) = 1$$

$$10k = 1$$

$$\therefore k = 0.1$$



3.2 Expectation and Variance

Result	Formula	Note
Expectation (mean)	$\mathrm{E}(X) = \mu = \sum x\,\mathrm{P}(X=x)$	long-run average
Expectation of a function	$\mathrm{E}(g(X)) = \sum g(x)\,\mathrm{P}(X=x)$	e.g. $\mathrm{E}(X^2) = \sum x^2 \mathrm{P}(X=x)$
Variance	$\mathrm{Var}(X) = \mathrm{E}[(X-\mu)^2] = \mathrm{E}(X^2) - [\mathrm{E}(X)]^2$	$\mathrm{Var}(X) \geqslant 0$
Standard deviation	$\sigma = \sqrt{\mathrm{Var}(X)}$	

GC: enter x in **L1**, $\mathrm{P}(X=x)$ in **L2**. **STAT** → **CALC** → **1-Var Stats L1,L2**. Then $\bar{x} = \mathrm{E}(X)$, $\sigma\mathbf{x} = \sigma$ (not **Sx**); $\mathrm{Var}(X) = \sigma^2$. $\mathrm{E}(X^2) = \mathrm{Var}(X) + [\mathrm{E}(X)]^2$, or read $\Sigma\mathbf{x}^2$.

$$\mathrm{E}(X) = 1(0.1) + 2(0.2) + 3(0.3) + 4(0.4) = 3$$

$$\mathrm{E}(X^2) = 1(0.1) + 4(0.2) + 9(0.3) + 16(0.4) = 10$$

$$\mathrm{Var}(X) = 10 - 3^2 = 1$$

3.3 Linear Functions and Combinations

Result	Formula	Condition
Linear function	$\mathrm{E}(aX+b)=a\mathrm{E}(X)+b$	always
	$\mathrm{Var}(aX+b)=a^2\mathrm{Var}(X)$	always; b does not change spread
Sum / difference	$\mathrm{E}(aX\pm bY)=a\mathrm{E}(X)\pm b\mathrm{E}(Y)$	always
	$\mathrm{Var}(aX\pm bY)=a^2\mathrm{Var}(X)+b^2\mathrm{Var}(Y)$	X, Y **independent**; variances **add** for $\pm$
Sum of n observations	$\mathrm{E}(X_1+X_2+\cdots+X_n)=n\mu$	always (independence not needed)
	$\mathrm{Var}(X_1+X_2+\cdots+X_n)=n\sigma^2$	$X_1,\ldots,X_n$ **independent** observations of X
One observation $\times n$	$\mathrm{E}(nX)=n\mu$	always
	$\mathrm{Var}(nX)=n^2\sigma^2$	always; $\mathrm{Var}(nX)\neq\mathrm{Var}(X_1+\cdots+X_n)$

Example 3.1

$Y=3X-2$, using $\mathrm{E}(X)=3$, $\mathrm{Var}(X)=1$

$$\mathrm{E}(Y)=3(3)-2=7$$

$$\mathrm{Var}(Y)=3^2(1)=9$$

Example 3.2

Prize W: 0, 4, 10 with probabilities 0.6, 0.3, 0.1; cost to play \$3

$$\mathrm{E}(W)=0+1.2+1=2.2$$

$$\mathrm{E}(\text{gain})=2.2-3=-0.8$$

$\therefore$ not a fair game

Note

- If a game is **fair**, $\mathrm{E}(\text{gain})=0$.
- To find the expected value of a variable other than the original X, there are two options:
  (1) link the new variable to the old one, e.g. $Y=5-2X$, then apply the expectation formulas;
  (2) draw an entirely new table for the new variable and find the expectation directly from that table.

4 Binomial Distribution

4.1 The Binomial Model

Item	Result	Note
Notation	$X \sim \mathrm{B}(n,p)$	X = no. of successes in n trials
Probability	$\mathrm{P}(X=r) = {}^{n}\mathrm{C}_{r}\, p^{r}(1-p)^{n-r}, \quad r=0,1,\ldots,n$	MF27
Mean	$\mathrm{E}(X) = np$	MF27 (no proof)
Variance	$\mathrm{Var}(X) = np(1-p)$	MF27 (no proof)

Conditions for $X \sim \mathrm{B}(n,p)$ (state in context)

- fixed number n of trials
- each trial has exactly 2 outcomes: success / failure
- $\mathrm{P}(\text{success}) = p$ is the **same** for every trial
- trials are **independent**

Note

$X \sim \mathrm{B}(n,p)$ counts the number of successes out of n trials. So $0 \leqslant X \leqslant n$, $X \in \mathbb{Z}$.

Model fails when …	Why
selecting without replacement from a small population	p changes; trials not independent
counting trials until the first success	n is not fixed
events are linked (e.g. family members, same day)	trials not independent
p varies (e.g. by time of day, by type of item)	p not constant

4.2 Using the GC

Probability	Conversion	GC command
$\mathrm{P}(X=r)$	not needed	`binompdf(n,p,r)`
$\mathrm{P}(X \leqslant r)$	not needed	`binomcdf(n,p,r)`
$\mathrm{P}(X<r)$	$\mathrm{P}(X \leqslant r-1)$	`binomcdf(n,p,r-1)`
$\mathrm{P}(X \geqslant r)$	$1-\mathrm{P}(X \leqslant r-1)$	`1-binomcdf(n,p,r-1)`
$\mathrm{P}(X>r)$	$1-\mathrm{P}(X \leqslant r)$	`1-binomcdf(n,p,r)`
$\mathrm{P}(a \leqslant X \leqslant b)$	$\mathrm{P}(X \leqslant b)-\mathrm{P}(X \leqslant a-1)$	`binomcdf(n,p,b)-binomcdf(n,p,a-1)`

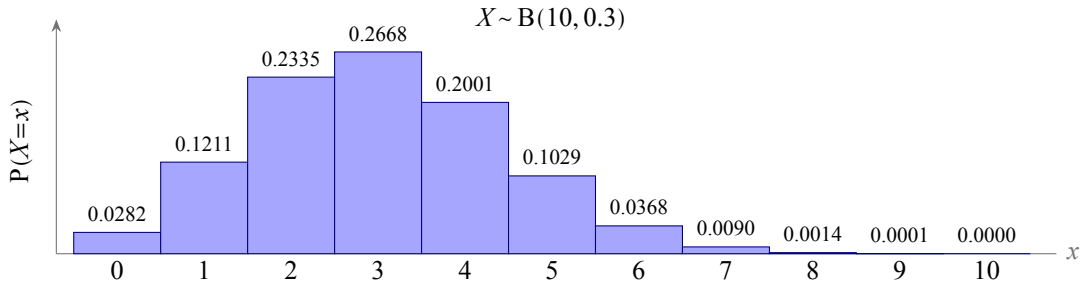
Note

- **binompdf(…)**: $P(X=r)$
- **binomcdf(…)**: $P(X \leqslant r)$

Any other inequalities must be converted to the above two forms before keying them into your GC.

GC: 2nd VARS (DISTR) → **binompdf(** / **binomcdf(**.

GC: solve for p or n: Y=, $Y_1 = $ **binomcdf(10,X,3)** (X plays p), $Y_2 =$ the given value; GRAPH; 2nd TRACE (CALC) → **5:intersect**. For n: use 2nd GRAPH (TABLE) instead.



Example 4.1

$X \sim \mathrm{B}(10, 0.3)$

$$\begin{aligned} \mathrm{P}(X=3) &= {}^{10}\mathrm{C}_3 (0.3)^3 (0.7)^7 \\ &= 0.267 \\ \mathrm{P}(X \geqslant 4) &= 1 - \mathrm{P}(X \leqslant 3) \\ &= 1 - 0.6496 = 0.350 \end{aligned}$$

$$\mathrm{E}(X) = 3, \ \mathrm{Var}(X) = 2.1$$

Example 4.2

$X \sim \mathrm{B}(20, 0.4)$. Find the probability that X is within 2 standard deviations of the mean.

$$\begin{aligned} \mu &= 20(0.4) = 8 \\ \sigma &= \sqrt{20(0.4)(0.6)} \\ &= \sqrt{4.8} = 2.1909 \\ \mu - 2\sigma &= 3.618 \\ \mu + 2\sigma &= 12.382 \\ \mathrm{P}(3.618 < X < 12.382) &= \mathrm{P}(4 \leqslant X \leqslant 12) \\ &= \mathrm{P}(X \leqslant 12) - \mathrm{P}(X \leqslant 3) \\ &= 0.97897 - 0.01596 \\ &= 0.963 \end{aligned}$$

Example 4.3

$X \sim \mathrm{B}(10, 0.3)$. Find the probability that the tenth result is the 3rd success.

$$\text{let } Y \sim \mathrm{B}(9, 0.3)$$

$$\mathrm{P}(Y=2) = {}^{9}\mathrm{C}_2 (0.3)^2 (0.7)^7$$

$$= 0.26683$$

$$\therefore \mathrm{P}(\text{10th is 3rd success}) = \mathrm{P}(Y=2) \times 0.3$$

$$= 0.26683 \times 0.3$$

$$= 0.0800$$

4.3 Common Question Types

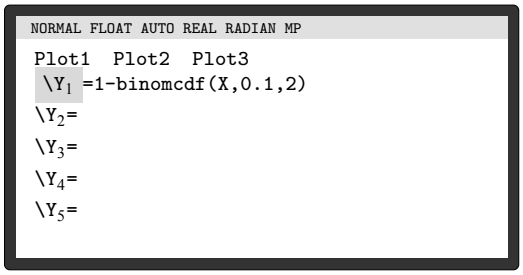
Type	Method
Smallest n with $\mathrm{P}(X \geqslant 1) > c$	Let $X \sim \mathrm{B}(n, p)$ (p is given), then use **GC table** to solve for n.
Unknown p	form the equation in p (e.g. $\mathrm{P}(X=0) = (1-p)^n$); solve; or use GC graph to solve for p
Given a condition	conditional probability: $\mathrm{P}(X=r \mid X \geqslant 1) = \dfrac{\mathrm{P}(X=r)}{\mathrm{P}(X \geqslant 1)}$, $r \geqslant 1$
Most probable value (mode) r	**Compare mode r with its directly adjacent neighbours** • $\mathrm{P}(X=r) > \mathrm{P}(X=r+1)$ • $\mathrm{P}(X=r) > \mathrm{P}(X=r-1)$ Equivalently, solve $\dfrac{\mathrm{P}(X=r)}{\mathrm{P}(X=r+1)} > 1$
Trial r is the k-th success	first $r-1$ trials contain $k-1$ successes, then a success: ${}^{r-1}\mathrm{C}_{k-1} p^{k-1} (1-p)^{r-k} \times p$ Alternatively, $\underbrace{{}^{r-1}\mathrm{C}_{k-1} p^{k-1} (1-p)^{r-k}}_{\text{define } Y \sim \mathrm{B}(r-1, p), \text{ and } \mathrm{P}(Y=k-1)} \times p$
Total / mean of N independent copies of $X \sim \mathrm{B}(n, p)$ (N is large)	Use the CLT (Ch 6) with $\mu = np$, $\sigma^2 = np(1-p)$ • $\bar{X} \sim \mathrm{N}\left(\mu, \dfrac{\sigma^2}{N}\right) = \mathrm{N}\left(np, \dfrac{np(1-p)}{N}\right)$ approximately by CLT • $X_1 + X_2 + \cdots + X_N \sim \mathrm{N}(N\mu, N\sigma^2) = \mathrm{N}(Nnp, Nnp(1-p))$ approximately by CLT For use of CLT, usually coupled with the phrases "estimate the probability" or "by using a suitable approximation".
Estimate p from data	$p \approx \dfrac{\text{total successes}}{\text{total trials}} = \dfrac{\bar{x}}{n}$ (frequency table: $\sum xf / n \sum f$)
Sketch of $\mathrm{B}(n, p)$	vertical bars at $r = 0, 1, \ldots, n$; label each probability above the bars; skewed right if $p < 0.5$, left if $p > 0.5$, symmetric if $p = 0.5$
Double binomial	new p' from the first part, e.g. $Y \sim \mathrm{B}(m, p')$ where $p' = \mathrm{P}(X \geqslant 2)$ ("on how many of m days")

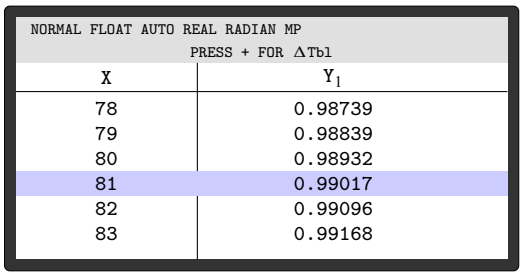
Example 4.4

$X \sim \mathrm{B}(n, 0.1)$. Find the smallest value of n such that $\mathrm{P}(X \geqslant 3) > 0.99$.

$$\mathrm{P}(X \geqslant 3) > 0.99$$

$$1 - \mathrm{P}(X \leqslant 2) > 0.99$$





X	Y_1
78	0.98739
79	0.98839
80	0.98932
81	0.99017
82	0.99096
83	0.99168

$$\therefore n_{\min} = 81$$

Example 4.5

$X \sim \mathrm{B}(10, p)$. Given that the mode of X is 3, find the range of p.

$$\mathrm{P}(X=3) > \mathrm{P}(X=2)$$

$${}^{10}\mathrm{C}_3 p^3 (1-p)^7 > {}^{10}\mathrm{C}_2 p^2 (1-p)^8$$

$$120p > 45(1-p)$$

$$165p > 45$$

$$p > \frac{3}{11}$$

$$\mathrm{P}(X=3) > \mathrm{P}(X=4)$$

$${}^{10}\mathrm{C}_3 p^3 (1-p)^7 > {}^{10}\mathrm{C}_4 p^4 (1-p)^6$$

$$120(1-p) > 210p$$

$$120 > 330p$$

$$p < \frac{4}{11}$$

$$\therefore \frac{3}{11} < p < \frac{4}{11}$$

Example 4.6

The number of defective items in a box of 20 is modelled by $X \sim \mathrm{B}(20, 0.3)$. A random sample of 50 boxes is taken. By using a suitable approximation, estimate the probability that the mean number of defective items per box exceeds 6.2.

$$\mathrm{E}(X) = 20(0.3) = 6$$

$$\mathrm{Var}(X) = 20(0.3)(0.7) = 4.2$$

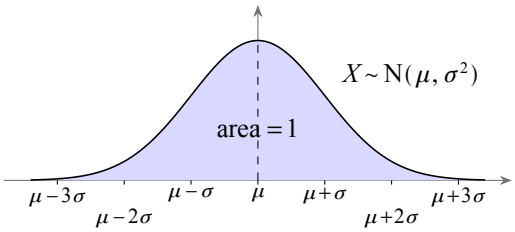
$\because N = 50$ is large,

$$\bar{X} \sim \mathrm{N}\left(6, \frac{4.2}{50}\right) \text{ approximately by CLT}$$

$$\therefore \mathrm{P}(\bar{X} > 6.2) = 0.245$$

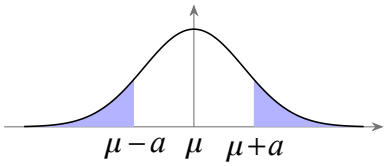
5 Normal Distribution

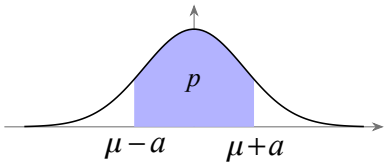
5.1 Properties



- bell-shaped, symmetric about $x=\mu$
- total area = 1; mean = μ; variance = σ^2
- continuous: $\mathrm{P}(X=a)=0$, so $\mathrm{P}(X<a)=\mathrm{P}(X\leqslant a)$
- σ larger $\Rightarrow$ wider and flatter

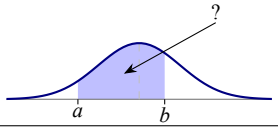
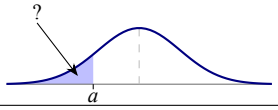
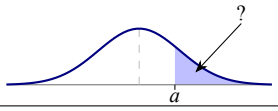
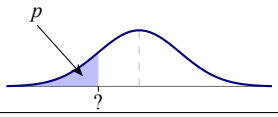
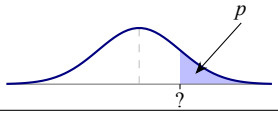
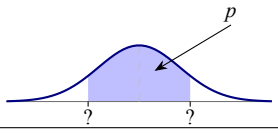
Item	Result	Note
Standardise	$Z=\dfrac{X-\mu}{\sigma}\sim \mathrm{N}(0,1)$	Z = number of s.d. from the mean
Symmetry	$\mathrm{P}(X<\mu-a)=\mathrm{P}(X>\mu+a)$	$\mathrm{P}(Z<-z)=\mathrm{P}(Z>z)=1-\mathrm{P}(Z<z)$
Complement	$\mathrm{P}(X>a)=1-\mathrm{P}(X<a)$	total area is 1





Left: equal tails. Right: $\mathrm{P}(\mu-a<X<\mu+a)=p$, so each tail is $\frac{1-p}{2}$.

5.2 Finding Probabilities and Values (GC)

Find	GC command	Graph
$P(a<X<b)$	`normalcdf(a,b,μ,σ)`	
$P(X<a)$	`normalcdf(-1E99,a,μ,σ)`	
$P(X>a)$	`normalcdf(a,1E99,μ,σ)`	
x with $P(X<x)=p$	`invNorm(p,μ,σ,LEFT)`	
x with $P(X>x)=p$	`invNorm(p,μ,σ,RIGHT)`	
x with $P(\mu-c<X<\mu+c)=p$	`invNorm(p,μ,σ,CENTER)` the two values $(\mu \pm c)$ need to be symmetrical about μ	

Note

Remember to square root your variance before keying into GC!

Example 5.1

$X \sim \mathrm{N}(60, 8^2)$: find $\mathrm{P}(52 < X < 70)$, and x with $\mathrm{P}(X > x) = 0.2$

$$\mathrm{P}(52 < X < 70) = 0.736$$

$$\mathrm{P}(X < x) = 0.8$$

$$\therefore x = 66.7$$

Example 5.2

Find μ, σ given $\mathrm{P}(X < 30) = 0.1$ and $\mathrm{P}(X > 46) = 0.2$

Let $X \sim \mathrm{N}(\mu, \sigma^2)$

$$\mathrm{P}(X < 30) = 0.1, \quad \mathrm{P}(X > 46) = 0.2$$

$$\mathrm{P}\left(Z < \tfrac{30-\mu}{\sigma}\right) = 0.1, \quad \mathrm{P}\left(Z > \tfrac{46-\mu}{\sigma}\right) = 0.2$$

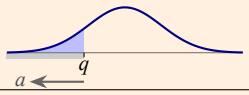
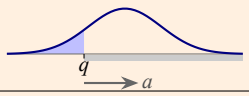
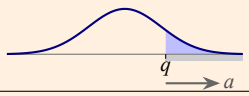
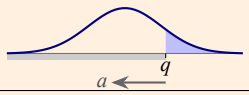
$$\frac{30-\mu}{\sigma} = -1.2816, \quad \frac{46-\mu}{\sigma} = 0.8416$$

Solving simultaneously,

$$\sigma = 7.54, \quad \mu = 39.7$$

Sometimes if the probabilities given are similar, e.g. $\mathrm{P}(X < 30) = 0.1$ & $\mathrm{P}(X > 50) = 0.1$, we should use symmetry to find μ, i.e. $\mu = \frac{30+50}{2} = 40$.

! invNorm Double Inequalities

Inequality	Solution	Graph	Note
$\mathrm{P}(X \leqslant a) < k$	$a <$ `invNorm(k,μ,σ,LEFT)`		keep the outer sign <
$\mathrm{P}(X \leqslant a) > k$	$a >$ `invNorm(k,μ,σ,LEFT)`		keep the outer sign >
$\mathrm{P}(X \geqslant a) < k$	$a >$ `invNorm(k,μ,σ,RIGHT)`		flip the outer sign <
$\mathrm{P}(X \geqslant a) > k$	$a <$ `invNorm(k,μ,σ,RIGHT)`		flip the outer sign >

5.3 Linear Functions and Combinations of Normals

Situation	Distribution	Condition
$aX+b$	$\mathrm{N}(a\mu+b, a^2\sigma^2)$	X normal
$aX \pm bY$	$\mathrm{N}(a\mu_X \pm b\mu_Y, a^2\sigma_X^2+b^2\sigma_Y^2)$	X, Y independent normals
$X_1+X_2+\cdots+X_n$	$\mathrm{N}(n\mu, n\sigma^2)$	independent copies of X
nX	$\mathrm{N}(n\mu, n^2\sigma^2)$	one observation × n

Common mistake

- Variances **add** for both $X+Y$ and $X-Y$: $\text{Var}(X-Y)=\text{Var}(X)+\text{Var}(Y)$.
- $\text{Var}(2X)=4\text{Var}(X)\neq\text{Var}(X_1+X_2)=2\text{Var}(X)$.

Example 5.3

$X\sim\text{N}(120,15^2)$, $Y\sim\text{N}(100,8^2)$ independent; $\text{P}(X>Y)$

$$X-Y\sim\text{N}(20,15^2+8^2)$$
$$=\text{N}(20,17^2)$$
$$\text{P}(X>Y)=\text{P}(X-Y>0)$$
$$=0.880$$

5.4 Further Question Types

Type	Method
Percentage change ("loses 5%")	new variable is $0.95X$: $\text{N}(0.95\mu,0.95^2\sigma^2)$ (not $X-5$)
Total of different items (e.g. 2 of X and 3 of Y)	$T=X_1+X_2+Y_1+Y_2+Y_3$: $\text{E}(T)=2\mu_X+3\mu_Y$, $\text{Var}(T)=2\sigma_X^2+3\sigma_Y^2$; with scaled items add $a^2\sigma^2$
"differs by less than c"	$\text{P}(\lvert X-Y\rvert<c)=\text{P}(-c<X-Y<c)$: `normalcdf(-c,c,μ,σ)`
Unknown μ or σ (one condition)	standardise: $\frac{a-\mu}{\sigma}=$ `invNorm(p)`, solve (the sign of z follows the tail)
Range of σ (or μ) for $\text{P}(X>a)<p$	$\frac{a-\mu}{\sigma}>$ `invNorm(1-p)`; $\sigma>0$, so multiply through without reversing; then solve
Normal event, then count ("k of n items")	$p=\text{P}(\text{event})$ by GC; then $\text{B}(n,p)$
Sketch of normal curves	bell shape, axis marked at μ, $\mu\pm\sigma$, $\mu\pm2\sigma$, $\mu\pm3\sigma$; same μ: larger σ is wider and lower; different μ: curves shifted; each has area 1
Model unsuitable	compute $\text{P}(X<0)$ (or outside a natural range): not negligible; or non-symmetric data

Example 5.4

$X\sim\text{N}(50,3^2)$, $Y\sim\text{N}(30,2^2)$ independent; $T=2X+3Y$, find $\text{P}(T>200)$

$$\text{E}(T)=2(50)+3(30)=190$$
$$\text{Var}(T)=4(9)+9(4)=72$$
$$T\sim\text{N}(190,72)$$
$$\text{P}(T>200)=0.119$$

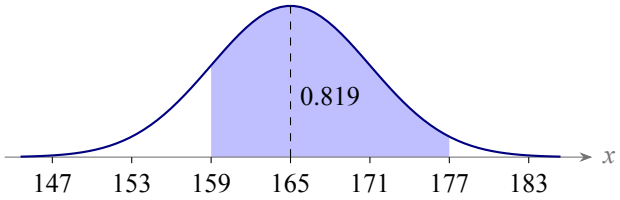
Example 5.5

The mass, X grams, of a bag of rice is modelled by $X\sim\text{N}(500,12^2)$. A bag is underweight if its mass is less than 480 grams. 20 bags are chosen at random. Find the probability that at least 2 of the bags are underweight.

$$X\sim\text{N}(500,12^2)$$
$$\text{P}(X<480)=0.04779$$
$$\text{let } Y\sim\text{B}(20,0.04779)$$
$$\text{P}(Y\geqslant2)=1-\text{P}(Y\leqslant1)$$
$$=1-0.75249$$
$$=0.248$$

Example 5.6

The height, X cm, of a student is modelled by $X \sim \mathrm{N}(165, 6^2)$. Sketch the normal curve of X, marking $\mu \pm \sigma$, $\mu \pm 2\sigma$ and $\mu \pm 3\sigma$ on the horizontal axis, and shade the region representing $\mathrm{P}(159 < X < 177)$. Hence find $\mathrm{P}(159 < X < 177)$.



$$\begin{aligned}\mathrm{P}(159 < X < 177) &= \mathrm{P}(-1 < Z < 2) \\ &= 0.819\end{aligned}$$

6 Sampling and the Central Limit Theorem

6.1 Population and Sample

Term	Meaning	Notation
Population	the whole group of interest	mean μ, variance σ^2 (parameters)
Sample	a part of the population, size n	$\bar{x}$, s^2 (statistics)
Random sample	• every member of the population is equally likely to be chosen • the selection is independent **Benefits:** avoid bias, and ensure the sample is representative of the population. **How to obtain:** number every member of the population with a distinct positive integer from 1 to N. Use a computer to randomly generate n distinct positive integers. Select the members whose numbers correspond to those generated by the computer.	

Not random if …	Reason
chosen by convenience (e.g. first 30 in the canteen, volunteers)	some members are more likely than others
only part of the population can be chosen	the rest has no chance
one choice affects another (e.g. friends chosen together)	not independent

6.2 Unbiased Estimates

Estimate of	Raw data	Coded: $\Sigma(x-a)$, $\Sigma(x-a)^2$
Mean μ	$\bar{x}=\dfrac{\sum x}{n}$	$\bar{x}=a+\dfrac{\sum(x-a)}{n}$
Variance σ^2	$s^2=\dfrac{1}{n-1}\left[\sum x^2-\dfrac{\left(\sum x\right)^2}{n}\right]$ or $s^2=\dfrac{1}{n-1}\sum(x-\bar{x})^2$	$s^2=\dfrac{1}{n-1}\left[\sum(x-a)^2-\dfrac{\left(\sum(x-a)\right)^2}{n}\right]$

GC: STAT → CALC → 1-Var Stats. Use Sx (=s, divides by $n-1$), so $s^2=$ Sx^2. For a frequency table, enter the values in L1 and the frequencies in L2, then 1-Var Stats L1,L2. Do **not** use σx (divides by n; this is not the unbiased estimate).

Example 6.1

$n=10, \sum x=125, \sum x^2=1830$

$$\bar{x}=\frac{125}{10}=12.5$$

$$s^2=\frac{1}{9}\left[1830-\frac{125^2}{10}\right]$$

$$=\frac{267.5}{9}=29.7$$

Example 6.2

given $\sum(x-10)=25, \sum(x-10)^2=330$

$$\bar{x}=10+\frac{25}{10}=12.5$$

$$s^2=\frac{1}{9}\left[330-\frac{25^2}{10}\right]$$

$$=\frac{267.5}{9}=29.7$$

Unbiased estimate of population variance

$$s^2=\frac{n}{n-1}\sigma_x^2$$

where $\sigma_x^2=\frac{1}{n}\sum(x-\bar{x})^2$ is the **sample variance** (GC: $\sigma\mathtt{x}^2$).

Example 6.3

(Hypothesis testing) [VJC 2025 Prelim P2 Q12]

A hospital introduces a new medication regimen to help patients recover from a viral infection more quickly. Previously, the average recovery time was 7.2 days.

…

It was discovered that an intern at the hospital had made an error in recording the recovery times. A random sample of another 30 patients was taken and their recovery times under the new regimen was taken and recorded by a senior researcher. The sample standard deviation was found to be 1.7 days and the sample mean time was denoted by k.

(f) A test at the 1% significance level concludes that the average recovery time did not improve from 7.2 days. Find the set of values of k. [3]

$$s^2=\frac{n}{n-1}(\text{sample variance})$$

$$=\frac{30}{29}(1.7)^2$$

$$=2.9897$$

Example 6.4

The masses, in kg, of a random sample of 5 parcels are 4, 7, 8, 10, 11. Find unbiased estimates of the population mean and variance.

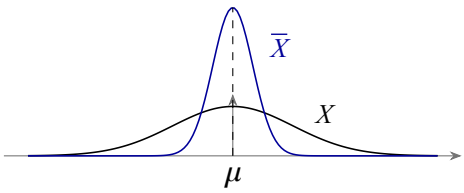
$$\bar{x} = \frac{4+7+8+10+11}{5} = 8$$

$$\sum (x-\bar{x})^2 = (-4)^2+(-1)^2+0^2+2^2+3^2$$
$$= 30$$

$$s^2 = \frac{1}{n-1}\sum (x-\bar{x})^2$$
$$= \frac{30}{4}$$
$$= 7.5$$

6.3 Distribution of the Sample Mean

Population	Distribution of $\overline{X}$	Reason
Normal, $X \sim \mathrm{N}(\mu, \sigma^2)$	$\overline{X} \sim \mathrm{N}(\mu, \frac{\sigma^2}{n})$ (exact)	any n; linear combination of normals
Not normal / unknown, n **large** (e.g. $n \geqslant 30$)	$\overline{X} \sim \mathrm{N}(\mu, \frac{\sigma^2}{n})$ approximately by CLT	**Central Limit Theorem**



Same centre μ. $\overline{X}$ is **narrower**: s.d. $\frac{\sigma}{\sqrt{n}}$ (larger n, narrower).

Example 6.5

population mean 80, s.d. 12, $n = 40$

$$\overline{X} \sim \mathrm{N}(80, \frac{12^2}{40}) \text{ approximately by CLT}$$

$$\mathrm{P}(\overline{X} > 83) = 0.0569$$

Central Limit Theorem

For a random sample of size n (large, e.g. $n \geqslant 30$) from **any** population with mean μ and variance σ^2:

- $\overline{X} \sim \mathrm{N}(\mu, \frac{\sigma^2}{n})$ approximately by CLT.
- $X_1 + X_2 + \cdots + X_n \sim \mathrm{N}(n\mu, n\sigma^2)$ approximately by CLT.

6.4 Which Distribution?

The question …	Use	Key step
number of successes in n independent trials	$\mathrm{B}(n, p)$	state the conditions
one measurement, normal	$\mathrm{N}(\mu, \sigma^2)$	standardise / GC
total or difference of independent normals	$\mathrm{N}(\cdot, \cdot)$	add the **variances**
mean of n values, population normal	$\overline{X} \sim \mathrm{N}(\mu, \sigma^2/n)$	exact
mean of n values, population not normal, n large	$\overline{X} \sim \mathrm{N}(\mu, \sigma^2/n)$ approximately by CLT	say "by CLT"

6.5 Further Question Types

Type	Method
Population given by a table or $\mathrm{B}(n, p)$, n observations large	$\mu = \mathrm{E}(X)$, $\sigma^2 = \mathrm{Var}(X)$ first (Ch 3, 4); then $\overline{X} \sim \mathrm{N}(\mu, \sigma^2/n)$ approximately by CLT
Total of many independent counts (e.g. defects in 100 batches)	$S = X_1 + \cdots + X_n \sim \mathrm{N}(n\mu, n\sigma^2)$ approximately by CLT
Least n for $\mathrm{P}(\overline{X} < c) \geqslant p$	$\frac{c-\mu}{\sigma/\sqrt{n}} \geqslant$ `invNorm(p)`, solve for $\sqrt{n}$; or GC table: $\mathrm{Y}_1 =$ `normalcdf` with s.d. $\sigma/\sqrt{\mathrm{X}}$, X as n
Is the sample size big enough?	$n < 30$ and population not known to be normal: CLT **cannot** be used; $n \geqslant 30$: say "by CLT"

Example 6.6

The number of defective items, X, in a box of 12 items is modelled by $X \sim \mathrm{B}(12, 0.25)$. A random sample of 40 boxes is taken. Find the probability that the mean number of defective items per box exceeds 3.4.

$$\mathrm{E}(X) = 12(0.25) = 3$$

$$\mathrm{Var}(X) = 12(0.25)(0.75) = 2.25$$

$\because n = 40$ is large

$$\overline{X} \sim \mathrm{N}(3, \frac{2.25}{40})$$

approximately by CLT

$$\mathrm{P}(\overline{X} > 3.4) = 0.0458$$

Example 6.7

The number of faults, X, recorded by a sensor in an hour has the following probability distribution.

x	0	1	2	3
$\mathrm{P}(X=x)$	0.4	0.3	0.2	0.1

45 independent hourly readings are taken. Find the probability that the total number of faults recorded exceeds 50.

$$\mathrm{E}(X) = 0(0.4) + 1(0.3) + 2(0.2) + 3(0.1)$$

$$= 1$$

$$\mathrm{E}(X^2) = 0(0.4) + 1(0.3) + 4(0.2) + 9(0.1)$$

$$= 2$$

$$\mathrm{Var}(X) = 2 - 1^2 = 1$$

let $S = X_1 + X_2 + \cdots + X_{45}$

$\because n = 45$ is large

$$S \sim \mathrm{N}(45(1), 45(1))$$

approximately by CLT

$$S \sim \mathrm{N}(45, 45)$$

approximately by CLT

$$\mathrm{P}(S > 50) = 0.228$$

7 Hypothesis Testing

7.1 Terminology

Term	Meaning
Null hypothesis H_0	the claim assumed true: $H_0: \mu = \mu_0$
Alternative hypothesis H_1	what we look for evidence of: $\mu < \mu_0$, $\mu > \mu_0$ (1-tail) or $\mu \neq \mu_0$ (2-tail)
Test statistic	$\bar{X}$ (or Z), whose distribution is found assuming H_0 is true
Level of significance α	probability of rejecting H_0 when H_0 is true; given in the question
Critical region	values of the test statistic for which H_0 is rejected; boundary = critical value
p-value	probability, assuming H_0 true, of a test statistic at least as extreme as the one observed; smallest level at which H_0 is rejected

7.2 Formulating the Hypotheses

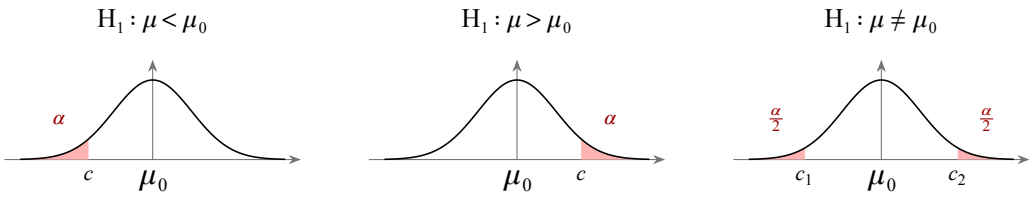
Wording	H_1	Type
decreased, less than, lower, overstated	$\mu < \mu_0$	1-tail (lower)
increased, more than, greater, understated	$\mu > \mu_0$	1-tail (upper)
changed, different, not correct, biased	$\mu \neq \mu_0$	2-tail

Define μ in context (e.g. "μ is the population mean mass"). H_0 always uses "=".

7.3 Test Statistic and Its Distribution

Situation	Distribution under H_0	Say
Normal population, σ^2 known	$\bar{X} \sim N(\mu_0, \frac{\sigma^2}{n})$	any n
Any population, n large, σ^2 known	$\bar{X} \sim N(\mu_0, \frac{\sigma^2}{n})$ approximately by CLT	"by CLT"
Any population, n large, σ^2 unknown	$\bar{X} \sim N(\mu_0, \frac{s^2}{n})$ approximately by CLT	"by CLT"; s^2 unbiased estimate

7.4 The Test: *p*-value and Critical Region



	Lower tail	**Upper tail**	**2-tail**
p-value	$P(\overline{X} \leqslant \bar{x})$	$P(\overline{X} \geqslant \bar{x})$	$2P(\overline{X} \leqslant \bar{x})$ if $\bar{x} < \mu_0$; $2P(\overline{X} \geqslant \bar{x})$ if $\bar{x} > \mu_0$
Reject H_0 if	*p*-value $\leqslant \alpha$		
Critical region	$\overline{X} \leqslant c$, $P(\overline{X} \leqslant c) = \alpha$	$\overline{X} \geqslant c$, $P(\overline{X} \geqslant c) = \alpha$	$\overline{X} \leqslant c_1$ or $\overline{X} \geqslant c_2$; each tail $\frac{\alpha}{2}$
Reject H_0 if	$\bar{x}$ lies in the critical region		

_p_-value of a two-tailed test

p-value (two-tailed test) = 2 × *p*-value (one-tailed test)

Steps of a test

1 $H_0: \mu = \mu_0$, $H_1: \mu \square \mu_0$; define μ; state α
2 distribution of $\overline{X}$ under H_0 (with reason: normal population / CLT)
3 *p*-value from the GC (or critical region)
4 compare: $p \leqslant \alpha \Rightarrow$ reject H_0; $p > \alpha \Rightarrow$ do not reject H_0
5 conclusion **in context**

Result	**Conclusion (in context)**
Reject H_0	there is sufficient evidence, at the α level of significance, to conclude that [H_1 in words]
Do not reject H_0	there is insufficient evidence, at the α level of significance, to conclude that [H_1 in words]

Never write "accept H_0" or "H_0 is true / proven".

GC: STAT → TESTS → 1:Z-Test. Inpt:Stats; enter $\mu 0$, σ (use *s* if σ unknown, large *n*), $\bar{x}$, n; choose $\neq \mu 0$, $< \mu 0$ or $> \mu 0$ to match H_1; Calculate gives *z* and *p*. With raw data use Inpt:Data with the list; for a frequency table first run 1-Var Stats L1,L2 to get $\bar{x}$ and Sx, then use Inpt:Stats.

Example 7.1

[N2016/H1/10]

A scientist claims that the mean top speed of cheetahs, in km/h, is 95. The top speed of each cheetah in a random sample of 40 cheetahs is recorded and the mean is found to be 96.3. It is known that the top speeds of cheetahs are normally distributed with standard deviation 4.1.

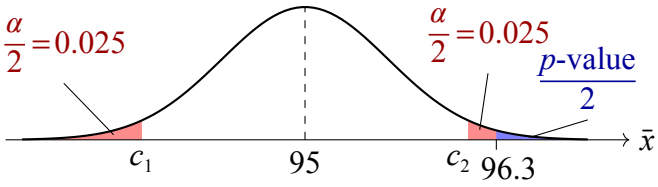
(i) Test the scientist's claim at the 5% significance level. [4]

Let X denote the top speed of a randomly selected cheetah in km/h, with population mean μ km/h.

$H_0: \mu = 95$
$H_1: \mu \neq 95$

Conduct a 2-tailed test at the 5% level of significance.

Test statistic:
Under H_0, *(since $n = 40$ is large, by central limit theorem,)*
$\overline{X} \sim N(95, \frac{4.1^2}{40})$ *approximately*



Reject H_0 if p-value $\leqslant$ 5% level of significance.

By GC, p-value = 0.0449

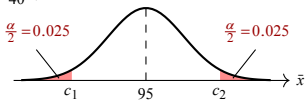
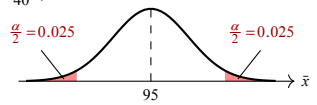
Thus, since p-value < 5%, we reject H_0 at the 5% level of significance. There is sufficient evidence to conclude that the mean top speed of cheetahs is not 95km/h. So the scientist's claim is not valid.

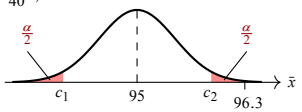
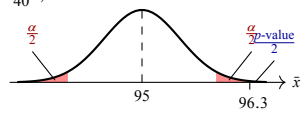
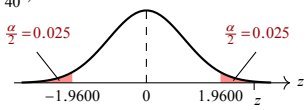
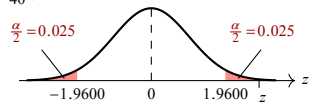
Common Definitions

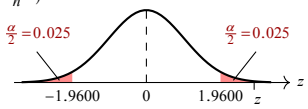
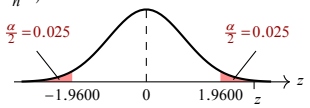
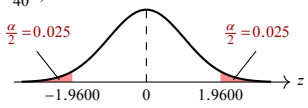
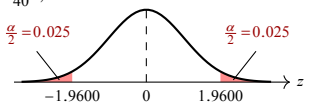
Term	Template	Definition (in context)
Random sample	every member of the population is equally likely to be chosen and the selections are independent.	each cheetah is equally likely to be chosen and the selections are independent
Level of significance	there is a probability of α% that we wrongly reject H_0, when H_0 is in fact true.	there is a probability of 5% that we wrongly conclude that the population mean top speed of cheetahs is not 95km/h when it is actually 95km/h.
p-value	it is the probability that the test statistic is as extreme or more extreme than the observed value given that H_0 is true. **OR** It is the smallest level of significance such that H_0 is rejected.	it is 2 times the probability that the observed sample mean top speed of cheetahs is more than 96.3km/h when the population mean top speed is 95km/h. **OR** It is the smallest level of significance to conclude that the mean top speed of cheetahs is not 95km/h.
Unbiased estimate	it is an estimate whose expected value is equal to the population parameter it is estimating.	

7.5 Reverse Questions

Reverse questions in hypothesis testing ask us to find the range of some variable, given that H_0 is rejected or H_0 is not rejected. Each question below uses the data of Example 7.1 [N2016/H1/10] with one quantity replaced by an unknown. In every case $H_0 : \mu = \mu_0$, $H_1 : \mu \neq \mu_0$ (2-tailed), and "the scientist's claim is not valid" means H_0 is rejected.

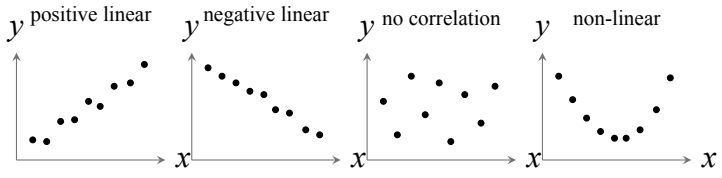
Index	**Question**	**Recommended method**	**Critical value method**	***p*-value method**
$\bar{x}$	A scientist claims that the mean top speed of cheetahs, in km/h, is 95. The top speed of each cheetah in a random sample of 40 cheetahs is recorded and the mean is found to be $\bar{x}$. It is known that the top speeds of cheetahs are normally distributed with standard deviation 4.1. A test is carried out at the 5% significance level. It is given that there is sufficient evidence that the scientist's claim is not valid. Find the set of possible values of $\bar{x}$.	Critical value $\overline{X} \sim \mathrm{N}(\checkmark, \checkmark)$	$H_0 : \mu = 95$ $H_1 : \mu \neq 95$ Test statistic: $\overline{X} \sim \mathrm{N}(95, \frac{4.1^2}{40})$  Since H_0 is rejected, by GC, $\bar{x} \leqslant 93.729$ or $\bar{x} \geqslant 96.271$ $\therefore \{\bar{x} \in \mathbb{R}^+ : \bar{x} \leqslant 93.729 \text{ or } \bar{x} \geqslant 96.271\}$	$H_0 : \mu = 95$ $H_1 : \mu \neq 95$ Test statistic: $\overline{X} \sim \mathrm{N}(95, \frac{4.1^2}{40})$  Reject H_0 if p-value $\leqslant 0.05$ If $\bar{x} > 95$: $2\mathrm{P}(\overline{X} \geqslant \bar{x}) \leqslant 0.05$ $\mathrm{P}(\overline{X} \geqslant \bar{x}) \leqslant 0.025$ by GC, $\bar{x} \geqslant 96.271$ If $\bar{x} < 95$: $2\mathrm{P}(\overline{X} \leqslant \bar{x}) \leqslant 0.05$ $\mathrm{P}(\overline{X} \leqslant \bar{x}) \leqslant 0.025$ by GC, $\bar{x} \leqslant 93.729$ $\therefore \{\bar{x} \in \mathbb{R}^+ : \bar{x} \leqslant 93.729 \text{ or } \bar{x} \geqslant 96.271\}$

Index	Question	Recommended method	Critical value method	*p*-value method
Level of significance $\alpha\%$	A scientist claims that the mean top speed of cheetahs, in km/h, is 95. The top speed of each cheetah in a random sample of 40 cheetahs is recorded and the mean is found to be 96.3. It is known that the top speeds of cheetahs are normally distributed with standard deviation 4.1. A test is carried out at the $\alpha\%$ significance level. It is given that there is sufficient evidence that the scientist's claim is not valid. Find the set of possible values of α.	*p*-value	$H_0: \mu = 95$ $H_1: \mu \neq 95$ Test statistic: $\overline{X} \sim N(95, \frac{4.1^2}{40})$  Since H_0 is rejected, 96.3 lies in the critical region: $96.3 \geqslant c_2$, where $P(\overline{X} \geqslant c_2) = \frac{\alpha}{200}$ $P(\overline{X} \geqslant 96.3) \leqslant P(\overline{X} \geqslant c_2)$ by GC, $0.022463 \leqslant \frac{\alpha}{200}$ $\alpha \geqslant 4.4926$ $\therefore \{\alpha \in \mathbb{R} : 4.4926 \leqslant \alpha < 100\}$	$H_0: \mu = 95$ $H_1: \mu \neq 95$ Test statistic: $\overline{X} \sim N(95, \frac{4.1^2}{40})$  Reject H_0 if *p*-value $\leqslant \frac{\alpha}{100}$ By GC, *p*-value $= 0.044926$ $0.044926 \leqslant \frac{\alpha}{100}$ $\alpha \geqslant 4.4926$ $\therefore \{\alpha \in \mathbb{R} : 4.4926 \leqslant \alpha < 100\}$
σ^2	A scientist claims that the mean top speed of cheetahs, in km/h, is 95. The top speed of each cheetah in a random sample of 40 cheetahs is recorded and the mean is found to be 96.3. It is known that the top speeds of cheetahs are normally distributed with variance σ^2. A test is carried out at the 5% significance level. It is given that there is sufficient evidence that the scientist's claim is not valid. Find the set of possible values of σ^2.	Critical value $\overline{X} \sim N(\checkmark, \frac{\square}{n})$	$H_0: \mu = 95$ $H_1: \mu \neq 95$ Test statistic: $\overline{X} \sim N(95, \frac{\sigma^2}{40})$  $z = \frac{96.3 - 95}{\sigma / \sqrt{40}} = \frac{1.3\sqrt{40}}{\sigma}$ $\because 96.3 > 95$, reject $H_0: z \geqslant 1.9600$ $\frac{1.3\sqrt{40}}{\sigma} \geqslant 1.9600$ $\sigma \leqslant \frac{1.3\sqrt{40}}{1.9600}$ $\sigma^2 \leqslant 17.597$ $\therefore \{\sigma^2 \in \mathbb{R} : 0 < \sigma^2 \leqslant 17.597\}$	$H_0: \mu = 95$ $H_1: \mu \neq 95$ Test statistic: $\overline{X} \sim N(95, \frac{\sigma^2}{40})$  Reject H_0 if *p*-value $\leqslant 0.05$ $2P(\overline{X} \geqslant 96.3) \leqslant 0.05$ $P(Z \geqslant \frac{1.3\sqrt{40}}{\sigma}) \leqslant 0.025$ $\frac{1.3\sqrt{40}}{\sigma} \geqslant 1.9600$ $\sigma \leqslant \frac{1.3\sqrt{40}}{1.9600}$ $\sigma^2 \leqslant 17.597$ $\therefore \{\sigma^2 \in \mathbb{R} : 0 < \sigma^2 \leqslant 17.597\}$

Index	Question	Recommended method	Critical value method	*p*-value method
n	A scientist claims that the mean top speed of cheetahs, in km/h, is 95. The top speed of each cheetah in a random sample of n cheetahs is recorded and the mean is found to be 96.3. It is known that the top speeds of cheetahs are normally distributed with standard deviation 4.1. A test is carried out at the 5% significance level. It is given that there is sufficient evidence that the scientist's claim is not valid. Find the set of possible values of n.	Critical value $\overline{X} \sim \mathrm{N}(\checkmark, \frac{\sigma^2}{\square})$	$\mathrm{H}_0: \mu = 95$ $\mathrm{H}_1: \mu \neq 95$ Test statistic: $\overline{X} \sim \mathrm{N}(95, \frac{4.1^2}{n})$  $z = \dfrac{96.3-95}{4.1/\sqrt{n}} = \dfrac{1.3\sqrt{n}}{4.1}$ $\because 96.3 > 95$, reject $\mathrm{H}_0: z \geqslant 1.9600$ $\dfrac{1.3\sqrt{n}}{4.1} \geqslant 1.9600$ $\sqrt{n} \geqslant \dfrac{1.9600(4.1)}{1.3} = 6.1815$ $n \geqslant 38.211$ $\therefore \{n \in \mathbb{Z}^+ : n \geqslant 39\}$	$\mathrm{H}_0: \mu = 95$ $\mathrm{H}_1: \mu \neq 95$ Test statistic: $\overline{X} \sim \mathrm{N}(95, \frac{4.1^2}{n})$  Reject H_0 if p-value $\leqslant 0.05$ $2\mathrm{P}(\overline{X} \geqslant 96.3) \leqslant 0.05$ $\mathrm{P}(Z \geqslant \frac{1.3\sqrt{n}}{4.1}) \leqslant 0.025$ $\dfrac{1.3\sqrt{n}}{4.1} \geqslant 1.9600$ $\sqrt{n} \geqslant 6.1815$ $n \geqslant 38.211$ $\therefore \{n \in \mathbb{Z}^+ : n \geqslant 39\}$
μ_0	A scientist claims that the mean top speed of cheetahs, in km/h, is μ_0. The top speed of each cheetah in a random sample of 40 cheetahs is recorded and the mean is found to be 96.3. It is known that the top speeds of cheetahs are normally distributed with standard deviation 4.1. A test is carried out at the 5% significance level. It is given that there is sufficient evidence that the scientist's claim is not valid. Find the set of possible values of μ_0.	Critical value $\overline{X} \sim \mathrm{N}(\blacksquare, \checkmark)$	$\mathrm{H}_0: \mu = \mu_0$ $\mathrm{H}_1: \mu \neq \mu_0$ Test statistic: $\overline{X} \sim \mathrm{N}(\mu_0, \frac{4.1^2}{40})$  $z = \dfrac{96.3-\mu_0}{4.1/\sqrt{40}}$ reject $\mathrm{H}_0: z \leqslant -1.9600$ or $z \geqslant 1.9600$ $96.3-\mu_0 \leqslant -1.9600\left(\frac{4.1}{\sqrt{40}}\right)$ or $96.3-\mu_0 \geqslant 1.9600\left(\frac{4.1}{\sqrt{40}}\right)$ $96.3-\mu_0 \leqslant -1.2706$ or $96.3-\mu_0 \geqslant 1.2706$ $\mu_0 \geqslant 97.571$ or $\mu_0 \leqslant 95.029$ $\therefore \{\mu_0 \in \mathbb{R}^+ : \mu_0 \leqslant 95.029 \text{ or } \mu_0 \geqslant 97.571\}$	$\mathrm{H}_0: \mu = \mu_0$ $\mathrm{H}_1: \mu \neq \mu_0$ Test statistic: $\overline{X} \sim \mathrm{N}(\mu_0, \frac{4.1^2}{40})$  Reject H_0 if p-value $\leqslant 0.05$ If $\mu_0 < 96.3$: $2\mathrm{P}(\overline{X} \geqslant 96.3) \leqslant 0.05$ $\mathrm{P}(Z \geqslant \frac{96.3-\mu_0}{4.1/\sqrt{40}}) \leqslant 0.025$ $\frac{96.3-\mu_0}{4.1/\sqrt{40}} \geqslant 1.9600$ $\mu_0 \leqslant 95.029$ If $\mu_0 > 96.3$: $2\mathrm{P}(\overline{X} \leqslant 96.3) \leqslant 0.05$ $\mathrm{P}(Z \leqslant \frac{96.3-\mu_0}{4.1/\sqrt{40}}) \leqslant 0.025$ $\frac{96.3-\mu_0}{4.1/\sqrt{40}} \leqslant -1.9600$ $\mu_0 \geqslant 97.571$ $\therefore \{\mu_0 \in \mathbb{R}^+ : \mu_0 \leqslant 95.029 \text{ or } \mu_0 \geqslant 97.571\}$

8 Correlation and Linear Regression

8.1 Scatter Diagrams



Item	Note
Independent x, dependent y	x: controlled / explains; y: the response (e.g. y = sales, x = advertising)
Describe a scatter diagram	direction (positive / negative), linear or curved, strength (close to a line), outliers
GC	enter x in `L1`, y in `L2`; `2nd` `Y=` (STAT PLOT) → `Plot1` on, scatter; `ZOOM` `9:ZoomStat`

8.2 Product Moment Correlation Coefficient r

Product Moment Correlation Coefficient (PMCC) (MF27)

$$r=\frac{\sum(x-\bar{x})(y-\bar{y})}{\sqrt{\sum(x-\bar{x})^2\sum(y-\bar{y})^2}}=\frac{S_{xy}}{\sqrt{S_{xx}S_{yy}}}$$

$$S_{xx}=\sum x^2-\frac{\left(\sum x\right)^2}{n},\quad S_{yy}=\sum y^2-\frac{\left(\sum y\right)^2}{n},\quad S_{xy}=\sum xy-\frac{\sum x\sum y}{n}$$

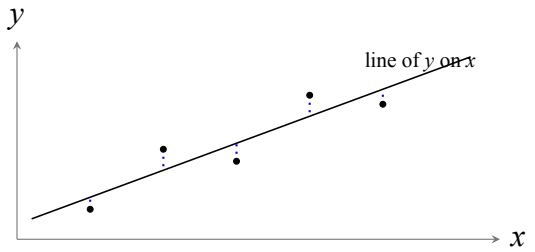
Value of r	Interpretation
$r=1$ or -1	all points lie exactly on a straight line (positive / negative gradient)
r close to 1	strong positive linear correlation
r close to -1	strong negative linear correlation
r close to 0	little or no **linear** correlation (a non-linear relation may still exist)

Note

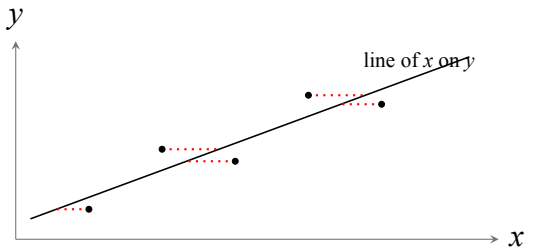
$-1\leqslant r\leqslant 1$. $|r|$ close to 1 only says a **linear** model fits well; check the scatter diagram. A strong correlation does **not** mean that x causes y.

GC: `2nd` `0` (CATALOG) → `DiagnosticOn` (once). Then `STAT` → `CALC` → `4:LinReg(ax+b) L1,L2` gives `a`, `b`, `r`. Use `8:LinReg(a+bx)` for $y=a+bx$.

8.3 Least-Squares Regression Line



Line of y on x minimises $\sum$(vertical distances)^2 (blue).



Line of x on y minimises $\sum$(horizontal distances)^2 (red).

	Estimate y given x	**Estimate x given y**
x is independent	y on x	y on x
y is independent (rare)	x on y	x on y
neither independent	y on x	x on y

Note

Both lines pass through $(\bar{x}, \bar{y})$.

Example 8.1

It is given that $n=6$, $\sum x=21$, $\sum y=35.4$, $\sum xy=143.4$, $\sum x^2=91$ and $\sum y^2=231$. Find the product moment correlation coefficient.

$$r=\frac{\sum xy-\frac{\sum x\sum y}{n}}{\sqrt{\left(\sum x^2-\frac{\left(\sum x\right)^2}{n}\right)\left(\sum y^2-\frac{\left(\sum y\right)^2}{n}\right)}}$$

$$=\frac{143.4-\frac{21(35.4)}{6}}{\sqrt{\left(91-\frac{21^2}{6}\right)\left(231-\frac{35.4^2}{6}\right)}}$$

$$=\frac{19.5}{\sqrt{17.5\times 22.14}}$$

$$\therefore r=0.991 \text{ (3 s.f.)}$$

8.4 Interpolation, Extrapolation and Interpretation

Item	Note
Interpolation	estimate for x **within** the data range: reliable if $\|r\|$ is close to 1 and the scatter diagram is linear
Extrapolation	estimate for x **outside** the data range: unreliable, the linear relation may not continue
Gradient b	the change in y per unit increase in x (in context)
Intercept a	value of y when $x=0$; meaningful only if $x=0$ is sensible in context
How well does the model fit?	$\|r\|$ close to 1 and points close to the line ⇒ good fit; comment on the range of x

Note

Reliability of an estimate requires **two** conditions:

- interpolation is used (state the data range);
- $|r|$ is close to 1, indicating a strong linear correlation between the two variables.

If any one of these fails, the estimate is unreliable.

8.5 Transformations to Achieve Linearity

Choose the model whose scatter diagram of (transformed variable, y) is the most linear, and whose $|r|$ is closest to 1.

Model	Scatter diagram	Plot	Shape ($a, b>0$)
$y=a+bx^2$		y against x^2	As x increases, y increases at an increasing rate.
$y=a+b\ln x$		y against $\ln x$	As x increases, y increases at a decreasing rate.
$y=a+\frac{b}{x}$		y against $\frac{1}{x}$	As x increases, y decreases at a decreasing rate.
$y=a\,e^{bx}$		$\ln y$ against x	As x increases, y increases at an increasing rate.

The shapes above are for $a, b>0$. In some cases a or b may be negative; the new shape can be analysed by plotting the graph in your GC.

Let w be the transformed variable (e.g. $w=x^2$, $\ln x$, $\frac{1}{x}$). Find r and the line of y on w with the GC (new list), then substitute $w=\ldots$ to estimate y. For $\ln y$ on x: $\ln y=a+bx \Rightarrow y=e^a e^{bx}$.

8.6 Sum of Squares of Residuals

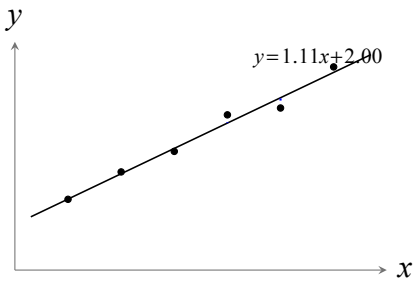
Residual

For a line $y=f(x)$ fitted to data, the residual at the data point (x_i, y_i) is

$$e_i = y_i - f(x_i) \qquad \text{(observed value − fitted value).}$$

Example 8.2

The least squares regression line for the 6 data points $(1, 3.1)$, $(2, 4.3)$, $(3, 5.2)$, $(4, 6.8)$, $(5, 7.1)$, $(6, 8.9)$ is $y=1.11x+2.00$. Find the residuals and the sum of squares of residuals.



$$e_1 = 3.1-(1.11(1)+2.00) = -0.01$$
$$e_2 = 4.3-(1.11(2)+2.00) = 0.08$$
$$e_3 = 5.2-(1.11(3)+2.00) = -0.13$$
$$e_4 = 6.8-(1.11(4)+2.00) = 0.36$$
$$e_5 = 7.1-(1.11(5)+2.00) = -0.45$$
$$e_6 = 8.9-(1.11(6)+2.00) = 0.24$$

$$\begin{aligned}\text{Sum of squares of residuals} &= (3.1-3.11)^2+(4.3-4.22)^2+(5.2-5.33)^2 \\ &\quad +(6.8-6.44)^2+(7.1-7.55)^2+(8.9-8.66)^2 \\ &= (-0.01)^2+(0.08)^2+(-0.13)^2+(0.36)^2+(-0.45)^2+(0.24)^2\end{aligned}$$

$$\therefore \sum e_i^2 = 0.4131$$

Note

Let c be the sum of squares of residuals of the least squares regression line. For any line $y=ax+b$,

$$\sum_{i=1}^{n}(y_i-(ax_i+b))^2 \geqslant c,$$

with equality if and only if $y=ax+b$ is the least squares regression line.

> **Note**
>
> **The residuals are squared** because some points are above the line, some below and some on the line. The residuals may be positive, negative or zero; adding them causes them to cancel out. Squaring them ensures they are all non-negative.
>
> **The least squares regression line is the line that minimises the sum of squares of residuals.**

8.7 Further Question Types

Type	Method
Residual and least squares	residual $e=y-(a+bx)$ (observed − predicted); positive above the line. For the least-squares line $\sum e=0$, so **squares** are used; the line minimising $\sum e^2$ is the best fit; smaller $\sum e^2 \Rightarrow$ better fit
Sum of squares of residuals	compute each e, square, add (or use the GC); compare two lines or two models by their $\sum e^2$
Missing data value / unknown	the line passes through $(\bar{x},\bar{y})$: $\bar{y}=a+b\bar{x}$ with the unknown in $\sum x$ or $\sum y$; solve
Extra point that leaves the line unchanged	the point $(\bar{x},\bar{y})$ (the mean point)
Change of units or coding, $x'=px+q$	r unchanged if $p>0$ (sign of r reverses if $p<0$); new line: substitute $x=\frac{x'-q}{p}$ into the fitted equation
Remove or change one point	recompute r, a, b on the GC; comment on the effect (e.g. $\lvert r\rvert$ nearer 1)
Choosing a model	scatter shape **in context** (e.g. levels off, cannot be negative), then $\lvert r\rvert$ closest to 1; state the constants and r for the chosen model
Estimate x from y (transformed model)	Use the fitted equation and substitute the given value in.
Sketch a scatter diagram	Label axes; plot every point (label $X_{\min}$, $X_{\max}$, $Y_{\min}$, $Y_{\max}$, dashed).

GC: `LinReg(a+bx) L1,L2,Y1` ([VARS] → Y-VARS → `1:Function` → `Y1` stores the line). In `L3` enter `L2-Y1(L1)`. Then [2nd] [STAT] (LIST) → MATH → `5:sum( L3²)` gives $\sum e^2$. Transformed variable: highlight the `L3` header and enter e.g. `ln(L1)`, `L1²` or `1/L1`, then run `LinReg` with `L3,L2`.

Example 8.3

x: 1, 2, 3, 4 y: 3, 6, 6, 10; line $y=2x+1$

$$e=0,\ 1,\ -1,\ 1$$

$$\sum e^2=0+1+1+1=3$$

least squares $y=1+2.1x$

$$\sum e^2=2.70<3$$

Example 8.4

$s=1.2t+5$, t in °C; use F in °F

$$t=\frac{5}{9}(F-32)$$

$$s=1.2\times\frac{5}{9}(F-32)+5$$

$$=\frac{2}{3}F-16.3$$